

Development of a wide range-operable, rich-lean low-NO_x combustor for NH₃ fuel gas-turbine power generation

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Abstract

Low-NO_x NH₃-air combustion power generation technology was developed by using a 50-kWe class micro gas-turbine system at the National Institute of Advanced Industrial Science and Technology (AIST), Japan, for the first time. Based on the global demand for carbon-free power generation as well as recent advances involving gas-turbine technologies, such as heat-regenerative cycles, rapid fuel mixing using strong swirling flows, and two-stage combustion with equivalence ratio control, we developed a low-NO_x NH₃-air non-premixed combustor for the gas-turbine system. Considering a previously performed numerical analysis, which proved that the NO reduction level depends on the equivalence ratio of the primary combustion zone in a NH₃-air swirl burner, an experimental study using a combustor test rig was carried out. Results showed that eliminating air flow through primary dilution holes moves the point of the lowest NO emissions to the lesser fuel flow rate. Based on findings derived by using a test rig, a rich-lean low NO_x combustor was newly manufactured for actual gas-turbine operations. As a result, the NH₃ single fueled low-NO_x combustion gas-turbine power generation using the rich-lean combustion concept succeeded over a wide range of power and rotational speeds, i.e., below 10–40 kWe and 75,000–80,000 rpm, respectively. The NO emissions were reduced to 337 ppm (16% O₂), which was about one-third of that of the base system. Simultaneously, unburnt NH₃ was reduced significantly, especially at the low electrical power output, which was indicative of

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the wider operating range with high combustion efficiency. In addition, N_2O emissions, which have a large Global Warming Potential (GWP) of 298, were reduced significantly, thus demonstrating the potential of NH_3 gas-turbine power generation with low environmental impacts.

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1. Introduction

For a long time, NH_3 was not considered a fuel because of its low burning velocity and high minimum ignition energy [1,2]. In the 1960s in the USA, gas-turbine engines that burned NH_3 gas had been researched, but the combustion efficiency was unacceptably low and thus the research and development on it was abandoned [3,4]. The recent demand for H_2 as an energy carrier has revived the interest in NH_3 as a fuel, as NH_3 fuel can serve as an H_2 energy carrier as well as a carbon-free fuel that can be used for power generation. Evans [5] proposed an NH_3 combustion gas turbine that uses cracked NH_3 . Valera et al. [6,7] tested both CH_4 - NH_3 and H_2 - NH_3 gas-turbine burners. In Japan, the Energy Carriers program is ongoing, in which advanced NH_3 production and utilization technologies are being developed.

During the last symposium, we reported that the National Institute of Advanced Industrial Science and Technology (AIST), Japan, successfully realized NH_3 -kerosene co-fired gas-turbine power generation in 2014 and NH_3 combustion gas-turbine power generation in 2015 by using a 50-kWe class micro gas turbine [8,9]. A micro gas turbine uses a regenerative heat exchanger to improve the thermal efficiency of the gas-turbine cycle. The regenerative heat exchanger increases the combustor inlet temperature (CIT) such that a high CIT reduces unburnt NH_3 emissions and improves the combustion efficiency [9]. The drawback of such technology is that it requires a large-size selective catalytic reduction (SCR) to decrease the high concentrations of NO_x . To increase the scale and number of NH_3 combustion gas-turbine power plants, it is necessary to develop low- NO_x combustion technology because large scale SCR and large numbers of SCR systems are essential without the development of low NO_x gas-turbine combustors. Present constraints include building costs and operational costs. These costs become more of an issue when the scale of gas turbines increases to middle and large sizes. Since NH_3 is typically known as a source of fuel NO and hence is used as a representative additive in studies of fuel NO chemistry from conventional fuels [10–12], achievement of low NO emissions from NH_3 flames may be challenging.

In a previous paper [9], we showed that NO and unburnt NH_3 emissions from NH_3 -air combustion at the original model combustor outlet depended

on the CIT (Fig. 8 in ref. [9]). For example, the NO emissions for a CIT of 580°C were 1050 ppm (at 16% O_2). Japanese governmental regulations stipulate the use of a 16% O_2 reference [13], although it is 15% in other countries. It is important to clarify the number of referenced O_2 percentages, not the values themselves. That figure can be translated to Fig. 1, in which the abscissa axis consists of the electrical power output. This means that for rotating speeds of 70,000, 75,000, and 80,000 rpm, NO and unburnt NH_3 emissions show one set of emission pattern repeated in three electrical power output ranges. It also shows that in the power generation system facility with SCR, the acceptable operating range for 75,000 rpm is from 26 to 39 kWe (see Fig. 7 in ref. [9]), which means that the lowest electrical power output is limited to where unburnt NH_3 does not exceed NO emissions before SCR. The NO and unburnt NH_3 emissions need to be minimized to both reduce the facility size with SCR and to extend the operation range of the electrical power output.

As for NO emission characteristics of NH_3 -air combustion, Hayakawa et al. [14] experimentally showed that when using an NH_3 -air premixed model swirl burner, the amount of NO and

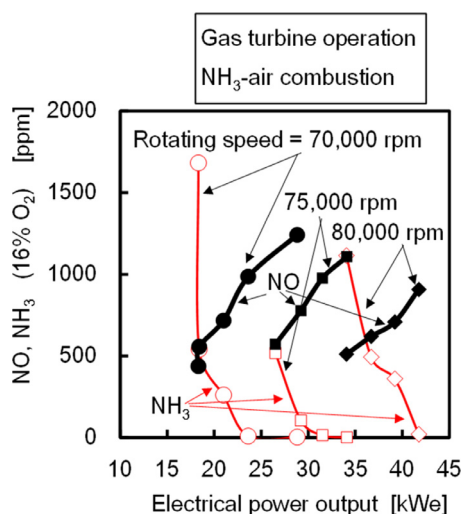


Fig. 1. Operation range limited by emissions, where unburnt NH_3 should not exceed NO .

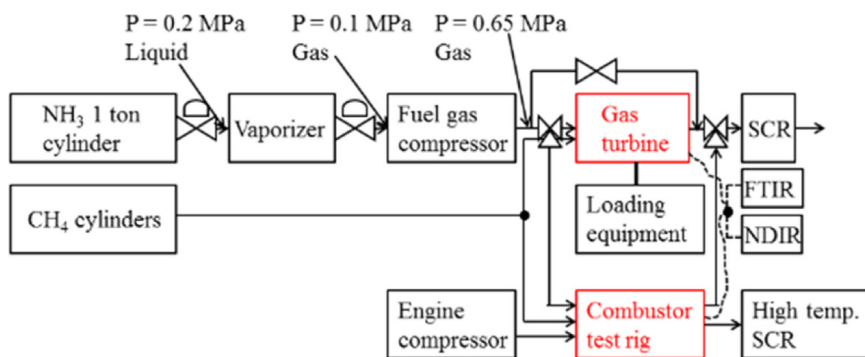


Fig. 2. Components of the combustor test rig.

unburnt NH_3 emissions will depend on the equivalence ratio, in which the former is at the minimum at the fuel rich side, with a value of 1.1. Hayakawa et al. [14] also carried out a one-dimensional flame simulation and compared the kinetic mechanisms to the works of Tian et al. [15], Miller et al. [16], Lindstedt and Selim [17], and Konnov and Ruyck [18]. By using LES numerical analysis, Kunkuma et al. [19] showed that a turbulent NH_3 -air premixed swirl combustor with secondary air injection has the potential to decrease NO emissions. In the Energy Carriers program, AIST has begun developing low- NO_x combustors by using a combustor test rig and micro gas-turbine system based on the research by Hayakawa et al. [14] and Kunkuma et al. [19], who are members of the program.

The purpose of this paper is to present the development of the low- NO_x gas-turbine combustor based on the rich-lean two-stage combustion concept for the first time, which is expected to result in low fuel NO formation in the primary zone and combustion of the produced H_2 and dilution by air in the secondary zone. The newly manufactured low- NO_x combustor was finally installed into the micro gas turbine, thus leading to successful operations and power generation that realized low- NO_x emissions, including N_2O , as well as low emissions of unburnt NH_3 .

2. Experimental facilities

Figure 2 shows the components of the combustor test rig. There are two parallel options for operating either a combustor test rig or a gas turbine. The combustor test rig was built close to the common NH_3 fuel supply facility. It consists of a combustor and a regenerative heat exchanger, and it was designed to investigate a combustor of the same size (length = 238 mm, diameter = 128 mm) with the same fuel and air flow rate as the case of the micro gas turbine [9]. In gas turbines, operating pressure and air flow rates change in various conditions, such as rotating

speeds and thermal power inputs. The combustor test rig has an orifice flow measurement system. Although it does not include a compressor or turbine, the combustor test rig is equipped with two middle-size engine compressors (Hokuetsu, PDSG750VRSD-4C5, Dry-air, $P = 0.5\text{--}1.4$ MPa, $Q_{\text{air}} = 21.2\text{--}25.5$ m³/min) and a high-temperature SCR; thus, the experimental results derived from the combustor test rig are expected to be applicable to the actual micro gas-turbine system.

For the gas-turbine operation experiments, the micro gas turbine and peripheral equipment in AIST that were used in the previous study were employed. Specifications of the facility have been mentioned elsewhere [8,9].

3. Results and discussion

3.1. Combustor configuration and emission characteristics in a test rig operation

Our previous studies of low NO_x emissions from NH_3 -air gas-turbine like burners [14,19,20] indicated that by adopting both two-stage combustion and control of the equivalence ratio of the primary combustion zone to around 1.1–1.2, substantial decreases in NO_x emissions were possible. Therefore, modifications of the base combustor liner were systematically applied to the test rig operations. The equivalence ratio in the primary zone was determined by the air flow rate of the flow paths, which interacted with each other.

Table 1 and Fig. 3 show the modifications applied to the combustor liners. Case 0 is the base combustor. Cases 1–5 are without the primary dilution holes. Case 1 and case 2 are when the areas of the secondary dilution holes were modified to $1.4 \times$ (multiple) and $2.0 \times$ (double) the initial, respectively. These cases are thought to have decreased the air flow rate and increased the equivalence ratio of the primary combustion zone. Case 3 is when the area of the air swirler

Table 1
Configuration of the combustor.

	Primary dilution holes	Area of air swirler	Sleeve-fitting gap area	Small cooling holes	Area of secondary dilution holes	Fuel/ Oxidizer
Case 0	With	1	With	With	1	NH ₃ / Air
Case 1	Without	1	With	With	1.4	NH ₃ / Air
Case 2	Without	1	With	With	2	NH ₃ / Air
Case 3	Without	0.5	With	With	1.4	NH ₃ / Air
Case 4	Without	1	With	Without	2	NH ₃ / Air
Case 5	Without	0.5	Without	Without	2	CH ₄ + NH ₃ /Air
xinlet of case0 in Eq. (1)	-	0.193	0.115	0.135	0.557	

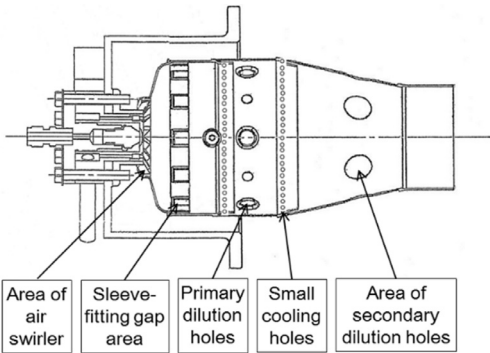


Fig. 3. Modifications applied to the base combustor.

was halved and the area of the secondary dilution hole was modified to $1.4 \times$. Case 4 is when small cooling holes were absent and the area of the secondary dilution hole was doubled. Case 5 is when the area of the air swirler was halved, without the sleeve-fitting gap area and without the small cooling holes, whereas the area of the secondary dilution hole was doubled.

Figure 4 shows the NO emission results of the test rig for NH₃-air combustion. The bottom abscissa axis represents the thermal power input (kWth). The top abscissa axis corresponds to the overall equivalence ratio of NH₃ (+CH₄)-air combustion because the air flow rates were fixed (1256 m³/h). The air flow rate of 1256 m³/h corresponded to the condition of 75,000 rpm rotation speed of the micro gas turbine. The pressure in the combustion chamber ranged from 257 to 304 kPa (abs) in Cases 0–5. The overall equivalence ratio was low because the turbine inlet temperature was up to 950 °C, which is common for a micro gas turbine. The size of the symbols represents the error bar heights. This applies to the subsequent figures as well. In case 1, the lowest NO emissions were observed at the point higher than the highest thermal power input. In case 2, the lowest NO emissions were observed at a thermal power input of 140 kWth, which was near the lowest thermal power input. In case 3, the thermal power input

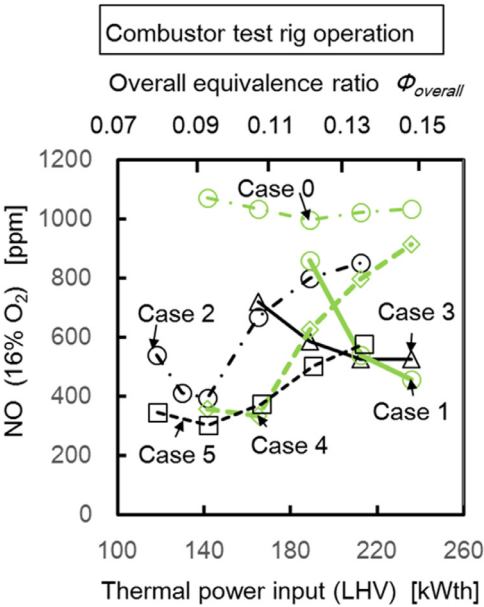


Fig. 4. NO emissions of the combustor test rig.

where the lowest NO emissions were observed was lower than that in case 1. In case 4, the lowest NO emissions were 334 ppm (16% O₂), which is a value that was 58 ppm lower than that in case 2. In case 5, the lowest NO emissions were 303 ppm (16% O₂), which is a value that was 31 ppm lower than that in case 4. Although case 5 included a CH₄ mixture, in the previous study [9] it was shown that the addition of NH₃ in the CH₄ + NH₃ mixture increased the NO emissions for the most part. These results indicate that closing the primary dilution holes, halving the area of the swirler, and doubling the area of the second dilution holes significantly influences the thermal power input, at which time the lowest NO emissions can be observed; cases without the sleeve-fitting gap area and without the small cooling holes had insignificant effects on the thermal power input.

Figure 5(a) shows the NO emission results from the test rig for NH₃-air combustion versus the

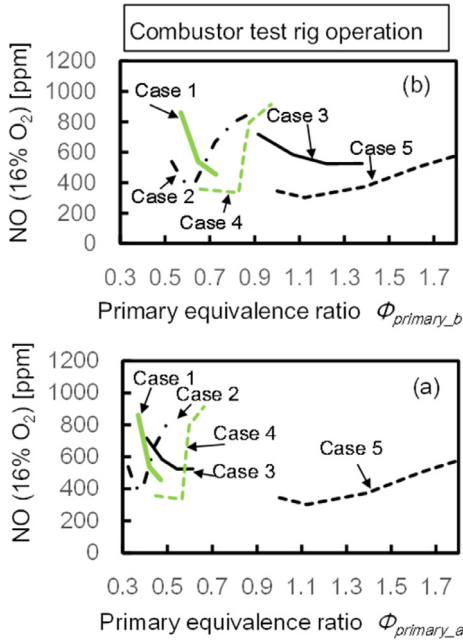


Fig. 5. Effect of $\Phi_{primary}$ on NO emissions of the combustor test rig; (a) Eq. (1) and (b) Eq. (1) without air flow through the sleeve-fitting gap and the small cooling holes.

equivalence ratio of the primary combustion zone. The equivalence ratio of the primary combustion zone was calculated as shown in Eq. (1).

$$\begin{aligned}\phi_{primary} &= \frac{3.57 \times \text{flow rate of NH}_3}{\text{Flow rate of air into the primary zone}} \\ &= \frac{3.57 \times \text{flow rate of NH}_3}{\text{Flow rate of total combustion air}} \\ &\quad \times \frac{1}{x_{inlet-i}} = \frac{\phi_{overall}}{x_{inlet-i}}\end{aligned}\quad (1)$$

where $\Phi_{primary}$ is the equivalence ratio of the primary combustion zone, $\Phi_{overall}$ is the overall equivalence ratio, and $x_{inlet-i}$ is the fraction of the total air that enters into the primary combustion zone as estimated by using the area of each flow path. $x_{inlet-i}$ is dependent on the size of the different air inlets on the combustor, and it was evaluated experimentally (Table 1) by measuring the air flow rate through the different air inlets using a laboratory combustor test rig [21].

The equivalence ratio of the primary combustion zone ($\Phi_{primary_a}$) where the lowest NO emissions were observed ranged from 0.4 to 1.1. This differs from the appropriate primary zone equivalence ratio of 1.1–1.2 that Hayakawa et al. [14] and Kunkuma et al. [19] had suggested. It is thought that the flow pattern differs because the base combustor adopts non-premixed combustion [9], whereas Kunkuma et al. [19] investigated premixed combustion. For non-premixed combustion, the air

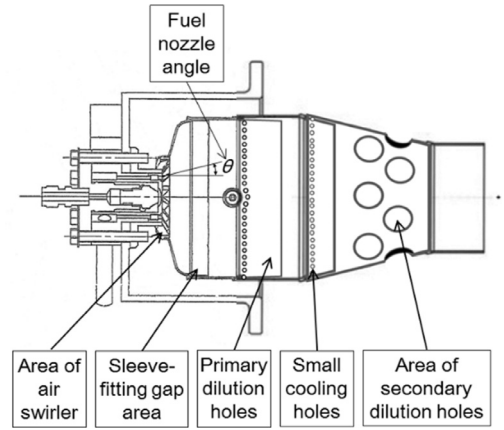


Fig. 6. Newly manufactured combustor.

flow near the wall may not be involved in mixing in the primary combustion zone; therefore, it does not affect the dilution of the mixture in the primary zone.

Figure 5(b) shows the NO emissions versus the primary zone equivalence ratio ($\Phi_{primary_b}$) based on the assumption of those effects. In calculating the primary zone equivalence ratio in Fig. 5(b), the air flow through the sleeve-fitting gap and the small cooling holes were not taken into account. The equivalence ratio of the primary combustion zone where the lowest NO emissions were observed changed to the range of 0.7–1.1.

Besides changing the combustor liners' design, we also investigated a change in the fuel nozzle shapes, in terms of variations in the injection angle from 0° to 45°, where 0° means parallel to the central axis of the combustor. The number and the diameter of the fuel nozzles were not changed. We observed that NO emissions for the 0° nozzle angle were the worst, while those for the 45° nozzle angle were at a good level.

3.2. Emission characteristics of the newly manufactured combustor in a gas-turbine operation

Based on the rig test results stated in the previous section, the rich-lean combustor was newly designed and manufactured. Table 2 and Fig. 6 show the design parameters and structure of the manufactured combustor. The combustor had no primary dilution holes, a decreased swirler area, and no sleeve-fitting gap area; additionally, the air flowing through the small cooling holes was blocked from getting into the primary zone and rather directed to the secondary combustion zone. Furthermore, the area of the secondary dilution holes was increased. The numbers of secondary dilution holes were set to 6 (case A), 9 (case B), and 12 (case C, D, E). The designed primary zone equivalence

Table 2
Newly manufactured combustor design.

	Primary dilution holes	Area of air swirler	Sleeve-fitting gap area	Small cooling holes	Area of secondary dilution holes	Designed primary equivalence ratio	Fuel nozzle angle
Case A	Without	Decrease	Without	Air emitting to secondary combustion zone	6 holes	0.7	45°
Case B	Without	Decrease	Without	Air emitting to secondary combustion zone	9 holes	0.9	45°
Case C	Without	Decrease	Without	Air emitting to secondary combustion zone	12 holes	1.1	45°
Case D	Without	Decrease	Without	Air emitting to secondary combustion zone	12 holes	1.1	30°
Case E	Without	Decrease	Without	Air emitting to secondary combustion zone	12 holes	1.1	0°

ratios at the rated electrical power output were 0.7 (case A), 0.9 (case B), and 1.1 (case C, D, E).

Figures 7 and 8 show the NO and the unburnt NH₃ emissions at the outlet of the newly manufactured combustor for the NH₃-air combustion gas-turbine operation. In Figs. 7 and 8, the abscissa axes are the generated electrical power output and CIT, respectively. The corresponding pressures in the combustion chamber ranged from 305 to 325 kPa (abs) (75,000 rpm) and 343 to 363 kPa (abs) (80,000 rpm). In Fig. 7, the lowest NO emissions were obtained in case B and case C when the fuel nozzle shapes were 45° nozzle angles. In case D and case E when the fuel nozzle shapes were 30° and 0°, respectively, the level of lowest emissions increased. This means that a large fuel nozzle angle will decrease the NO emissions. The NO emissions from the newly manufactured, rich-lean combustor decreased to one-third of that from the base model combustor. In addition, in case E, the unburnt NH₃ increased up to 685 ppm (16% O₂) at an electrical power output of 13 kWe, whereas in cases A to D, the unburnt NH₃ was below 55 ppm (16% O₂) at an electrical power output of 10 kWe. This shows that there were significant decreases when compared to the unburnt NH₃ emissions of Fig. 1. Therefore, the new combustor enabled a wider operating range of electrical power output and higher combustion efficiency. The rich-lean combustion for the NH₃-air system may have enhanced the combustion temperature, thus resulting in the decrease of unburnt NH₃.

As seen in Fig. 8, although the previous study [9] showed that the NO and unburnt NH₃ strongly depended on the CIT (case 0), the dependencies of the newly manufactured, rich-lean combustor differed in each case. Similar to the previous study [9], the influence of rotating speeds disappeared in view of the influence of the CIT in each case. It was also

shown that a CIT lower than 500 °C still suppressed the unburnt NH₃, thus indicating that the unburnt NH₃ of the NH₃-air combustion can be reduced in the gas turbine without a regenerative heat exchanger.

Figure 9 shows the highest temperature of the combustor liner obtained by thermocouples placed on the liner. During the operation of the micro gas turbine, combustor liner temperatures below 1000 °C are preferred. In case B and case C, the highest combustor liner temperature reached 1000 °C before reaching the highest electrical power output. Although adopting a large fuel nozzle angle decreases NO emissions, it also decreases the operating range because of the heat-resistance of the combustor liner. It is estimated that increasing the fuel nozzle angle positions the tip of the fuel mixture close to the combustor liner.

Figure 10 shows the thermal efficiencies. In the case of the 75,000 rpm rotating speed, the data were superimposed. In the case of the 80,000 rpm rotating speed, the thermal efficiencies were ranked as follows: case B > case C > case D > case E, but with small differences. Although the thermal efficiency of case 0 was higher than that of the others, the test was conducted with less pressure loss of exhaust gas through the SCR unit such that this case was not appropriate for comparisons. It has been shown that the configurations of rich-lean combustors do not affect the combustion efficiency, and almost the same thermal efficiency can be achieved as a base system using CH₄-air [9]. In fact, rich-lean combustion is widely used in coal combustion applications. In the case of hydrocarbon fuels, the intermediate product in the primary combustion zone is mainly CO, in which the oxidation rate is relatively low. In the case of NH₃ fuel, the product is mainly H₂, in which the oxidation rate is very high. Although the length of the combustor liner after

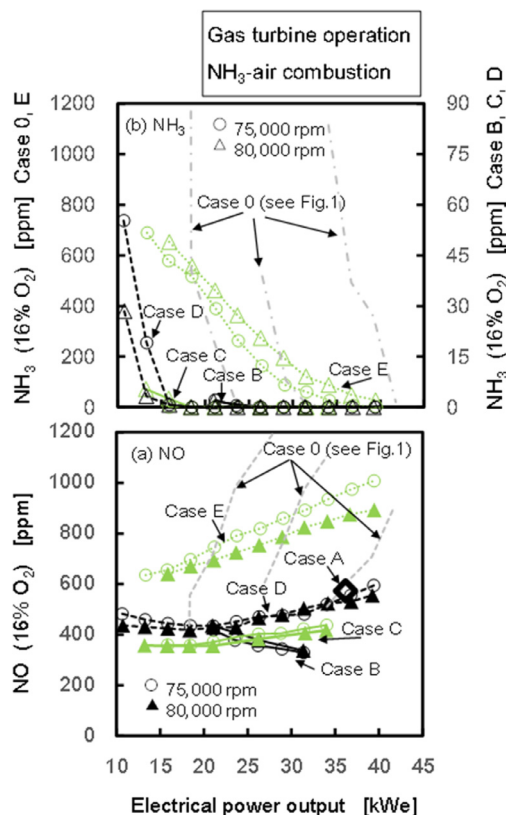


Fig. 7. Emissions of the newly manufactured combustor in a gas-turbine operation (electrical power output).

the secondary dilution holes was not extended, the high oxidation rate of H₂ can be expected to increase the combustion efficiency.

Figure 11 shows the N₂O emissions. The abscissa axis is the CIT. The Global Warming Potential (GWP) is the value of the effect of a greenhouse gas relative to that of CO₂, as announced by the Intergovernmental Panel on Climate Change (IPCC) for the Kyoto Protocol; the GWP of N₂O is 298 [22]. The GWP of CO₂ is 1; GWP is a dimensionless value. Thus, N₂O emissions need to be minimized. Figure 11 shows that decreases of the CIT increased the N₂O emissions. The rates of increase of N₂O emissions versus the CIT can be represented as follows: case E > case B > case D > case C. Again, the 0° nozzle angle gave the worst results. However, we were able to obtain very low N₂O emissions amounting to less than 4 ppm in most of the cases when the CIT was higher than 450 °C.

4. Conclusions

Low-NO_x NH₃-air combustion power generation technology was developed by using a 50-kWe class micro gas turbine and a combustor test rig fa-

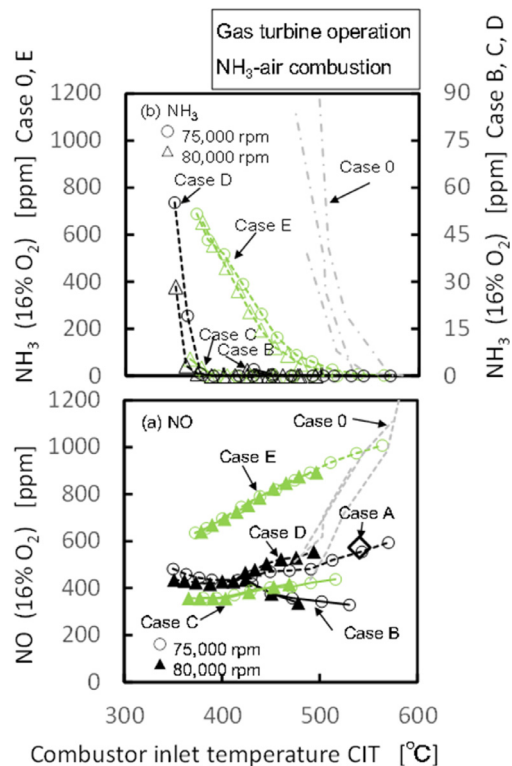


Fig. 8. Emissions of the newly manufactured combustor in a gas-turbine operation (combustor inlet temperature).

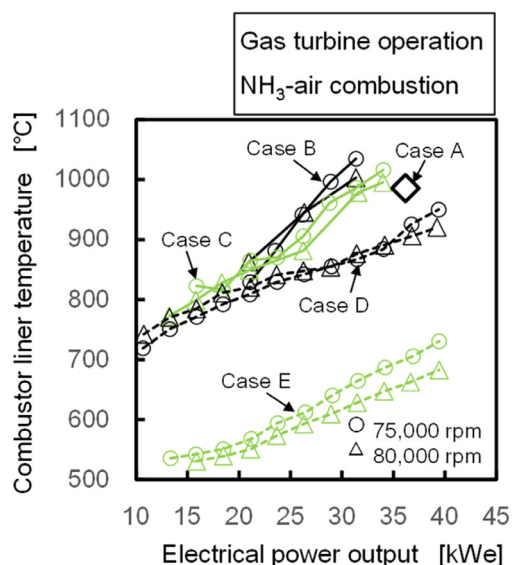


Fig. 9. Combustor liner temperature of the newly manufactured combustor in a gas-turbine operation.

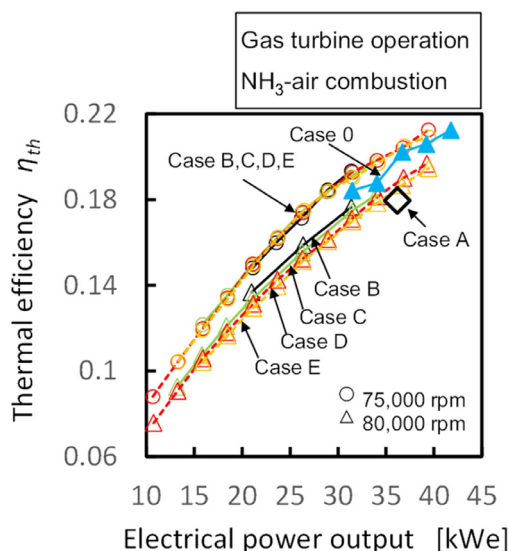


Fig. 10. Thermal efficiency of the newly manufactured combustor in a gas-turbine operation.

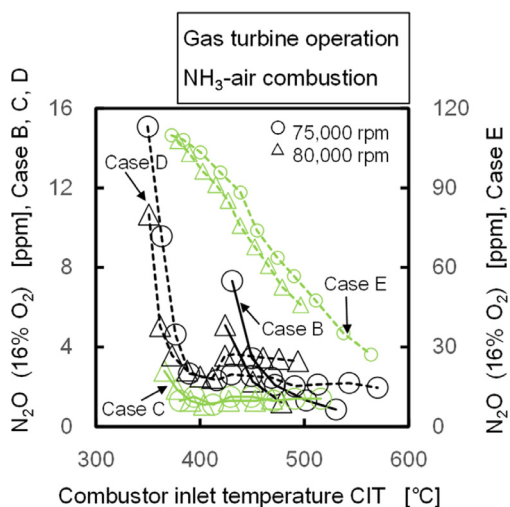


Fig. 11. N_2O emissions of the newly manufactured combustor in a gas-turbine operation.

cility. The rich-lean two-stage combustion concept was adopted for the non-premixed combustor to reduce NO formation in the primary combustion zone, enhance combustion of the H_2 produced, and dilute the introduced air in the secondary zone. The test rig experiments were first performed for tests of the modified base combustor. Then, based on the results, a rich-lean combustor for gas-turbine operations was newly manufactured and actual power generation was performed. The results obtained can be summarized as follows:

1. Reductions of NO_x emissions at the outlet of the modified combustor during the

test rig operation were successfully confirmed following the elimination of air flow into the primary zone, which indicates that rich-lean combustion is effective for NH_3 -air gas-turbine combustors.

2. The newly manufactured rich-lean combustor in actual gas-turbine operations could reduce NO emissions to 337 ppm minimum (16% O_2), which is about one-third of that of the base system, and it could simultaneously decrease unburnt NH_3 and N_2O emissions.
3. The rich-lean low- NO_x combustor developed in this study enabled a wider operating range of electrical power generation and rotational speeds, i.e., below 10–40 kWe and 75,000–80,000 rpm. Thus the new technology shows potential for being able to realize low- NO_x NH_3 single fueled power generation.

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References

- [1] F.J. Verkamp, M.C. Hardin, J.R. Williams, *Proc. Combust. Inst.* 11 (1967) 985–992.
- [2] A. Hayakawa, T. Goto, R. Mimoto, T. Kudo, H. Kobayashi, *Mech. Eng. J.* 2 (2015) 1–10. Paper No. 14-00402.
- [3] U.S. Army Engineer Research and Development Laboratories, Solar Division of International Harvester Company, Development of an ammonia-burning gas turbine engine, Final Technical Report, DA-44-009-AMC-824(T), Fort Belvoir, Solar, 1968, available at <<http://oai.dtic.mil/oai/oai?verb=getRecord&metadataPrefix=html&identifier=AD0671667>>.
- [4] D.T. Pratt, *Performance of Ammonia-Fired Gas-Turbine Combustors*, Berkley University of California, 1967 Technical Report No.9, DA-04-200-AMC-791(x) available at <http://www.dtic.mil/dtic/tr/fulltext/u2/657585.pdf>.
- [5] B. Evans, Using local green energy and ammonia to power gas turbine generators, *The 10th Annual NH3 Fuel Conference*, 2013.
- [6] M.A. Valera, R. Marsh, J. Runyon, D. Pugh, P. Beasley, T. Hughes, P. Bowen, *Appl. Energy* 185 (2017) 1362–1371.
- [7] M.A. Valera, D.G. Pugh, P. Marsh, G. Bulat, P. Bowen, *Int. J. Hydrog. Energy* 42 (2017) 24495–24503.
- [8] N. Iki, O. Kurata, T. Matsunuma, T. Inoue, M. Suzuki, T. Tsujimura, H. Furutani, Micro gas turbine firing kerosene and ammonia, ASME Turbo Expo 2015 GT2015-43689, 2015.

- [9] O. Kurata, N. Iki, T. Matsunuma, T. Inoue, T. Tsujimura, H. Furutani, H. Kobayashi, A. Hayakawa, *Proc. Combust. Inst.* 36 (2017) 3351–3359.
- [10] B.S. Haynes, *Combust. Flame* 28 (1977) 81–91.
- [11] C.J. Dasch, R.J. Blint, *Combust. Sci. Technol.* 41 (1984) 223–244.
- [12] N. Sullivan, A. Jensen, P. Glarborg, M.S. Day, J.F. Gracar, J.B. Bell, C.J. Pope, R.J. Kee, *Combust. Flame* 131 (2002) 285–298.
- [13] Ministry of the Environment, Japan, available at <<http://www.env.go.jp/air/osen/law/t-kise-6.html>> (in Japanese).
- [14] A. Hayakawa, Y. Arakawa, R. Mimoto, K.D. Kunkuma, A. Somarathne, T. Kudo, H. Kobayashi, *Int. J. Hydrog. Energy* 42 (2017) 14010–14018.
- [15] Z. Tian, Y. Li, L. Zhang, P. Glarborg, F. Qi, *Combust. Flame* 156 (2009) 1413–1426.
- [16] J.A. Miller, M.D. Smooke, R.M. Green, R.J. Kee, *Combust. Sci. Technol.* 34 (1983) 149–176.
- [17] R.P. Lindstedt, M.A. Selim, *Combust. Sci. Technol.* 99 (4–6) (1994) 277–298.
- [18] A.A. Konnov, J.D. Ruyck, *Combust. Sci. Technol.* 152 (2000) 23–37.
- [19] K.D. Kunkuma, A. Somarathne, S. Hatakeyama, A. Hayakawa, H. Kobayashi, *Int. J. Hydrog. Energy* 42 (2017) 27388–27399.
- [20] K.D. Kunkuma, A. Somarathne, A. Hayakawa, H. Kobayashi, *Bull. JSME, J. Fluid Sci. Technol.* 11 (2016) 1–10. Paper No.16-00126.
- [21] E.C. Okafor, K.D.K.A. Somarathne, A. Hayakawa, T. Kudo, O. Kurata, N. Iki, H. Kobayashi, *Proc. Combust. Inst.* (2018) 37. In Press.
- [22] G. Myhre, D. Shindell, F.-M. Bréon, et al., 2013: Anthropogenic and Natural Radiative Forcing, p.714, In: *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* [T.F. Stocker, D. Qin, G.-K. Plattner, M. Tignor, S.K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex and P.M. Midgley (eds.)], Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, available at <http://www.climatechange2013.org/images/report/WG1AR5_Chapter08_FINAL.pdf>.